

Fixed-multiplicity shuffles and balanced Smirnov words

The $e^{-(k-1)}$ law, Poisson adjacency statistics,
and explicit all-orders corrections

A research note on OEIS A007060, A193624, A330266, and related sequences

1 October 2026

Abstract

Let $B_{n,k}$ be the number of permutations of kn distinct cards, partitioned into n ranks of size k , in which neighboring cards never have the same rank. The OEIS entry A330266 [1] records the fixed- k conjecture (S. Kirgizov, 29 September 2023)

$$\frac{B_{n,k}}{(kn)!} \longrightarrow e^{-(k-1)}.$$

The case $k = 4$ was subsequently deduced on that entry from an approximation to a related sequence (D. Radcliffe, 9 September 2025), but the general statement remained posed there. We give a direct proof for every fixed $k \geq 2$ and prove considerably more. If $X_{n,k}$ is the number of equal-rank adjacencies in a uniformly random shuffle, then

$$X_{n,k} \xrightarrow{d} \text{Poisson}(k-1).$$

We derive an exact probability-generating integral whose kernel is an associated Laguerre polynomial, an exact inclusion-exclusion formula, a gamma-expectation representation, and a constructive expansion to arbitrary order. In particular, uniformly for fixed k ,

$$\begin{aligned} \frac{B_{n,k}}{(kn)!} = e^{-(k-1)} & \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2n^2} \right. \\ & \left. - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3n^3} + O_k(n^{-4}) \right]. \end{aligned}$$

The first four factorial cumulants are obtained in exact rational form; the mean is exactly $k-1$ and the variance is $(k-1) - (k-1)^2/(kn-1)$. For A330266 this gives

$$\frac{B_{n,4}}{(4n)!} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right).$$

We also treat the corresponding multiset sequences, local probabilities, Hamiltonian paths in complete multipartite graphs, ratio asymptotics, and a Lambert- W inverse transseries. Exact reproducibility code accompanies the article.

Scope and priority. The sequence definitions and historical comments below are taken from the OEIS pages cited in the bibliography. A330266 [1] now states that its particular $k = 4$ leading equivalent follows from an approximation to A190830. The point of the present note is the uniform fixed- k theorem, the Poisson law, exact factorial cumulants, and systematic correction terms. No claim of a complete literature search or historical priority is made; the results should be independently checked before submission or attribution.

Editorial note (1 October 2026, batch 73O2). Manuscript 57 of batch 73 of ProveIt’s incoming-reports intake, placed in commit 6e193dd4f; AI-assisted, unrefereed, not formalized. The write added the label prefix `bsw:`, citation commands, the names of the OEIS contributors, notes marked “[Added 1 October 2026, batch 73O2]”, Remark 6.3, one bibliography entry and Appendix C. The analytic tail step of Theorem 6.2 is a sketch; the fixed- k limit law does not depend on it.

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1 The OEIS problem and the family of sequences

Fix integers $n \geq 0$ and $k \geq 2$. There are n ranks, each containing k distinct cards. A shuffle is a permutation of all $N = kn$ cards. Let

$$X_{n,k} = \#\{1 \leq j < N : \text{the cards at positions } j \text{ and } j+1 \text{ have the same rank}\}. \quad (1)$$

Write

$$B_{n,k} = N! \mathbb{P}(X_{n,k} = 0), \quad p_{n,k} = \frac{B_{n,k}}{N!}. \quad (2)$$

If cards within one rank are made indistinguishable, let $S_{n,k}$ denote the number of admissible multiset words. Every rank word lifts to exactly $(k!)^n$ labeled permutations, so

$$B_{n,k} = (k!)^n S_{n,k}, \quad p_{n,k} = \frac{S_{n,k}}{(kn)!/(k!)^n}. \quad (3)$$

Thus the same probability governs the labeled and multiset versions.

The most visible OEIS rows are as follows.

k	labeled count $B_{n,k}$	interpretation	multiset count $S_{n,k}$
2	A007060 [2]	couples with no adjacent spouses	A114938 [4]
3	A193624 [3]	triples with no adjacent siblings	A193638 [5]
4	A330266 [1]	four cards in each rank	A321633 [6]

A330266 [1] records the exact integral

$$B_{n,4} = \int_0^\infty (t^4 - 12t^3 + 36t^2 - 24t)^n e^{-t} dt$$

and the conjectured fixed-multiplicity law, posed there by S. Kirgizov on 29 September 2023,

$$p_{n,k} \longrightarrow e^{-(k-1)}. \quad (4)$$

The entry also records a September 2025 deduction of the $k = 4$ equivalent from a separate approximation: D. Radcliffe (9 September 2025) combined A330266(n) = $24^n n!$ A190830(n) with V. Kotěšovec's approximation of A190830 and Stirling's formula. Our derivation below is direct, applies to every fixed k , and identifies the full limiting distribution behind (4).

1.1 Why a Poisson law is plausible but not automatic

For each of the $N - 1$ adjacent position pairs, the probability of equal ranks is $(k - 1)/(N - 1)$. Hence

$$\mathbb{E}X_{n,k} = (N - 1) \frac{k - 1}{N - 1} = k - 1. \quad (5)$$

If the rare adjacency indicators were independent, one would expect a Poisson limit of mean $k - 1$ and hence a zero probability $e^{-(k-1)}$. They are not independent: overlapping position pairs and the fixed rank multiplicities produce both local and global dependence. The proof therefore requires more than the heuristic “rare events with bounded mean.” The exact generating function will make the cancellation transparent.

1.2 Hamiltonian-path interpretation

Replace the deck by the complete n -partite graph with n parts of size k . A vertex ordering is an admissible shuffle precisely when consecutive vertices are joined. Consequently $B_{n,k}$ is the number of *oriented* Hamiltonian paths, counted as ordered vertex lists, in $K_{k,k,\dots,k}$. Unoriented paths are counted by $B_{n,k}/2$ when $kn > 1$. This connection places the problem among balanced Smirnov words and complete multipartite Hamiltonian enumeration [7, 9].

2 A weighted run transform

It is useful to enumerate all shuffles while recording the number of bad adjacencies. Define

$$A_{n,k}(u) = \sum_{\pi \in \mathfrak{S}_{kn}} u^{X_{n,k}(\pi)}, \quad G_{n,k}(u) = \frac{A_{n,k}(u)}{(kn)!} = \mathbb{E} u^{X_{n,k}}. \quad (6)$$

Thus $A_{n,k}(0) = B_{n,k}$ and $G_{n,k}(0) = p_{n,k}$.

Theorem 2.1 (Run-transform generating function). *For indeterminates $x_1, \dots, x_n$,*

$$\sum_w u^{X(w)} x_1^{|w|_1} \dots x_n^{|w|_n} = \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}, \quad (7)$$

where the sum is over all finite words on $\{1, \dots, n\}$, including the empty word. Therefore

$$A_{n,k}(u) = (k!)^n [x_1^k \dots x_n^k] \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}. \quad (8)$$

Proof. Compress every maximal run of equal letters to one letter. The compressed word is a Smirnov word, meaning that consecutive letters are distinct. A run of letter i and length $r \geq 1$ contributes $u^{r-1} x_i^r$, whose generating series is

$$y_i = \sum_{r \geq 1} u^{r-1} x_i^r = \frac{x_i}{1 - u x_i}.$$

The classical multivariate generating function of Smirnov words [10, 11] is

$$\frac{1}{1 - \sum_i y_i / (1 + y_i)}.$$

Indeed, it follows from the ordinary word series $1/(1 - \sum_i z_i)$ by replacing each nonempty run by one marked letter, or by the standard run-decomposition bijection. Substitution gives

$$\frac{y_i}{1 + y_i} = \frac{x_i}{1 + (1-u)x_i},$$

which proves (7). Extracting k occurrences of each rank gives multiset words; labeling the k copies independently in each rank contributes $(k!)^n$ and proves (8). $\square$

3 The exact Laguerre integral

Set $v = u - 1$ and define

$$P_{k,u}(t) = k! [x^k] \exp\left(\frac{tx}{1-vx}\right). \quad (9)$$

Theorem 3.1 (Exact integral and polynomial kernel). *For every $n \geq 0$, $k \geq 2$, and complex u ,*

$$A_{n,k}(u) = \int_0^\infty e^{-t} P_{k,u}(t)^n dt. \quad (10)$$

The kernel has the explicit forms

$$P_{k,u}(t) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{k!}{j!} t^j (u-1)^{k-j}, \quad (11)$$

$$= k!(u-1)^k L_k^{(-1)}\left(-\frac{t}{u-1}\right), \quad (12)$$

where the value at $u = 1$ is interpreted by continuity.

Proof. Use the coefficientwise Laplace identity

$$\frac{1}{1-s} = \int_0^\infty e^{-t} e^{ts} dt.$$

In (8), the exponential factors into n identical univariate series. Extracting x_i^k from each factor gives (10). Expanding $\exp(tx/(1-vx))$ first by the exponential degree and then extracting x^k gives (11). Finally, the standard expansion [12]

$$L_k^{(-1)}(z) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{(-z)^j}{j!}$$

gives (12). □

For no equal adjacencies, $u = 0$ and $v = -1$. Write

$$P_k(t) := P_{k,0}(t) = (-1)^k k! L_k^{(-1)}(t). \quad (13)$$

The first kernels are

$$\begin{aligned} P_2(t) &= t^2 - 2t, \\ P_3(t) &= t^3 - 6t^2 + 6t, \\ P_4(t) &= t^4 - 12t^3 + 36t^2 - 24t, \\ P_5(t) &= t^5 - 20t^4 + 120t^3 - 240t^2 + 120t. \end{aligned}$$

The third line recovers the integral printed on A330266, while the theorem explains it as one row of a uniform Laguerre family.

4 Exact inclusion–exclusion and a compact reciprocal polynomial

For $0 \leq m \leq k-1$, put

$$a_{k,m} = \binom{k-1}{m} \frac{k!}{(k-m)!} = m! \binom{k}{m} \binom{k-1}{m}, \quad (14)$$

and define the degree- $(k-1)$ polynomial

$$R_k(z) = \sum_{m=0}^{k-1} a_{k,m} z^m. \quad (15)$$

The no-adjacency reciprocal polynomial is

$$C_k(z) = z^k P_k(1/z) = R_k(-z). \quad (16)$$

Theorem 4.1 (Exact finite formula). *Let $N = kn$. Then*

$$B_{n,k} = \sum_{m_1, \dots, m_n=0}^{k-1} (-1)^M (N-M)! \prod_{i=1}^n a_{k,m_i}, \quad M = m_1 + \dots + m_n, \quad (17)$$

$$= \sum_{M=0}^{n(k-1)} (N-M)! [z^M] C_k(z)^n. \quad (18)$$

More generally, the factorial probability generating function

$$\Phi_{n,k}(v) := \mathbb{E}(1+v)^{X_{n,k}} = G_{n,k}(1+v) \quad (19)$$

satisfies the exact identity

$$\Phi_{n,k}(v) = \sum_{M=0}^{n(k-1)} \frac{[z^M] R_k(z)^n}{(N)_{\underline{M}}} v^M. \quad (20)$$

Proof. From (11),

$$z^k P_{k,u}(1/z) = \sum_{m=0}^{k-1} a_{k,m} v^m z^m = R_k(vz).$$

Raise to the n th power and integrate termwise in (10). Since $\int_0^\infty t^{N-M} e^{-t} dt = (N-M)!$, division by $N!$ yields (20). Setting $v = -1$ gives (17) and (18). $\square$

Formula (18) is a rook-polynomial form of inclusion–exclusion. It uses only a degree- $(k-1)$ polynomial raised to the n th power and is efficient for exact integer computation. In particular, for $k = 4$ it reproduces

$$B_{13,4} = 3668033946384704437729512814619767610579526911188666362431432294400,$$

the standard-deck value recorded on A330266.

5 The Poisson adjacency theorem

Recall that the r th falling factorial is $(x)_{\underline{r}} = x(x-1)\cdots(x-r+1)$.

Theorem 5.1 (Factorial moments). *For fixed $k \geq 2$ and fixed $r \geq 0$,*

$$\mathbb{E}(X_{n,k})_{\underline{r}} = (k-1)^r + O_{k,r}(n^{-1}). \quad (21)$$

Consequently,

$$\boxed{X_{n,k} \xrightarrow[n \rightarrow \infty]{d} \text{Poisson}(k-1).} \quad (22)$$

Proof. The factorial moment is $r!$ times the coefficient of v^r in (20). Since every term $a_{k,m}z^m$ in $R_k(z)$ has the same z -degree as the corresponding v -degree, the denominator for this coefficient is exactly $(N)_{\underline{r}}$. The leading contribution to $[z^r]R_k(z)^n$ selects $a_{k,1}z$ from r distinct factors and 1 from the rest:

$$[z^r]R_k(z)^n = \binom{n}{r} a_{k,1}^r + O_{k,r}(n^{r-1}).$$

Here $a_{k,1} = k(k-1)$. Therefore

$$\begin{aligned} \mathbb{E}(X_{n,k})_{\underline{r}} &= \frac{r! \binom{n}{r} [k(k-1)]^r + O_{k,r}(n^{r-1})}{(kn)_{\underline{r}}} \\ &= (k-1)^r + O_{k,r}(n^{-1}). \end{aligned}$$

These are the factorial moments of $\text{Poisson}(k-1)$. Ordinary moments are finite linear combinations of factorial moments, so every ordinary moment converges. The Poisson distribution is moment-determinate; the method of moments gives (22). $\square$

Corollary 5.2 (The fixed-multiplicity OEIS law). *For every fixed integer $k \geq 2$,*

$$\boxed{\frac{B_{n,k}}{(kn)!} = p_{n,k} \longrightarrow e^{-(k-1)}.} \quad (23)$$

Equivalently,

$$S_{n,k} \sim \frac{(kn)!}{(k!)^n} e^{-(k-1)}. \quad (24)$$

Proof. Because the variables are integer-valued,

$$\mathbb{P}(X_{n,k} = 0) = \mathbb{P}(-\tfrac{1}{2} < X_{n,k} < \tfrac{1}{2}).$$

The endpoints $\pm \frac{1}{2}$ carry no mass under the limiting Poisson law, so weak convergence implies convergence of this probability. Hence $\mathbb{P}(X_{n,k} = 0) \rightarrow \mathbb{P}(\text{Poisson}(k-1) = 0) = e^{-(k-1)}$. Equation (24) follows from (3). $\square$

Remark 5.3. The proof is short because the correct object is not merely the zero-event count. The entire adjacency statistic has a stable Poisson limit. The constant $k-1$ is already forced by the exact mean (5).

6 A gamma expectation and an all-orders expansion engine

The exact finite sum admits a useful probabilistic transform. Let T_N have the gamma density

$$f_N(t) = \frac{t^N e^{-t}}{N!}, \quad t > 0, \quad T_N \sim \text{Gamma}(N+1, 1). \quad (25)$$

Since

$$\mathbb{E} T_N^{-M} = \frac{(N-M)!}{N!} = \frac{1}{(N)_{\underline{M}}} \quad (0 \leq M \leq N), \quad (26)$$

we obtain the following exact formula.

Proposition 6.1 (Gamma representation). *For $N = kn$,*

$$\boxed{\Phi_{n,k}(v) = \mathbb{E} \left[R_k \left(\frac{v}{T_N} \right)^n \right]}. \quad (27)$$

The gamma variable is concentrated at scale $\sqrt{N}$ around $N+1$. Since $R_k(w) = 1 + k(k-1)w + O_k(w^2)$, the exponent in (27) tends to $(k-1)v$. This already supplies another proof of the Poisson limit. More importantly, it supports a complete Laplace expansion.

For a nonnegative integer M , define polynomials $H_r(M)$ by

$$\prod_{j=0}^{M-1} \frac{1}{1-jx} = \sum_{r \geq 0} H_r(M) x^r. \quad (28)$$

Thus $H_r(M) = h_r(0, 1, \dots, M-1)$, a complete homogeneous symmetric polynomial. The first four are

$$H_0(M) = 1, \quad (29)$$

$$H_1(M) = \frac{M(M-1)}{2}, \quad (30)$$

$$H_2(M) = \frac{M(M-1)(3M^2 + M - 2)}{24}, \quad (31)$$

$$H_3(M) = \frac{M^2(M-1)^2(M+1)(M+2)}{48}. \quad (32)$$

Let $D = v d/dv$.

Theorem 6.2 (Constructive all-orders expansion). *Fix $k \geq 2$. As $n \rightarrow \infty$, uniformly for v in every compact subset of $\mathbb{C}$, $\Phi_{n,k}(v)$ has a Poincare expansion in integer powers of $1/n$. A coefficientwise construction is*

$$\boxed{\Phi_{n,k}(v) \sim \sum_{r \geq 0} \frac{1}{N^r} H_r(D) \left[R_k \left(\frac{v}{N} \right)^n \right], \quad N = kn.} \quad (33)$$

After expanding the bracket in $1/N$, collecting total powers gives any desired correction term. In particular, every coefficient is an explicit polynomial in v and a rational function of k , multiplied by $e^{(k-1)v}$.

Proof. Write

$$R_k(z)^n = \sum_{M=0}^{n(k-1)} r_{n,M} z^M.$$

Then (20) is

$$\Phi_{n,k}(v) = \sum_M r_{n,M} \frac{v^M}{N^M} \frac{N^M}{(N)_{\underline{M}}}.$$

For fixed M ,

$$\frac{N^M}{(N)_{\underline{M}}} = \prod_{j=0}^{M-1} \left(1 - \frac{j}{N}\right)^{-1} \sim \sum_{r \geq 0} \frac{H_r(M)}{N^r}.$$

Since D acts on v^M by multiplication by M , this is precisely the formal operator rule (33).

It remains to justify the asymptotic use when M is not fixed. The exact gamma representation (27) does so. Split the integral into $|T_N - N| \leq N^{2/3}$ and its complement. On the upper tail, gamma Chernoff bounds are enough. On the lower tail one combines the polynomial bound $|R_k(v/t)| \leq C_{k,V}(1 + t^{1-k})$ for $|v| \leq V$ with Stirling's formula; the exact negative moments show that terms whose total polynomial degree is proportional to N are in fact superexponentially small. Thus the weighted complement is smaller than every fixed power of N^{-1} . On the central interval, T_N^{-1} , $\log R_k(v/T_N)$, and the gamma density all possess Taylor expansions with remainders uniform for v in a prescribed compact set. After the change $T_N = N + \sqrt{N}y$, each coefficient is a polynomial in y times a Gaussian majorant, so termwise integration is valid to arbitrary fixed order. Odd half-powers cancel in the integrated expansion, leaving integer powers of $1/N$. The coefficientwise operator computation and the Laplace computation agree because both expand the same exact negative moments (26). This proves the Poincare expansion and the stated uniformity. $\square$

Remark 6.3 (Status of the analytic step). [Added 1 October 2026, batch 73O2.] The tail estimate is sketched. The coefficientwise operator rule is exact algebra, but the passage to an asymptotic statement uniform for v in compact sets is not proved in detail: the gamma Chernoff bounds, the superexponential smallness of terms of total degree proportional to N , and the cancellation of odd half-powers are asserted, not derived. A uniform factorial-moment bound of the type proved in the companion path-forest report **a189281-path-forest-expansions** of this collection (its Lemma 4.1, “Uniform factorial-moment bound”, label **spf:lem:moment-bound**, which bounds the k th factorial moment by $(\theta e^2)^k/k!$ for its own model) would make the truncation rigorous; that lemma is not proved here for the present model.

Accordingly, Theorem 6.2 and everything that uses it rest on this sketched step: the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ stated after Theorem 7.1, Theorem 8.1 (through “the fifth and higher factorial cumulants begin at $O_k(n^{-4})$ ”), Corollaries 8.2 and 8.3, the specializations (44)–(47), the count, ratio and inverse expansions (49), (52) and (54), and the correction terms in the proposed OEIS comments. They do not depend on it: Theorems 2.1, 3.1, 4.1, 5.1 and 7.1, Proposition 6.1, and Corollary 5.2, the fixed- k limit law that settles the conjecture on A330266. The intake's numerical check is consistent with the displayed coefficients: for $k = 2, \dots, 5$ the residual after the n^{-3} term of (42), multiplied by $e^{k-1}n^4$, is nearly the same at $n = 100$ and $n = 400$, as an $O(n^{-4})$ remainder requires. That is evidence, not a proof.

7 Exact factorial cumulants

Define factorial cumulants $\kappa_r^{(F)}$ by

$$\log \Phi_{n,k}(v) = \sum_{r \geq 1} \kappa_r^{(F)} \frac{v^r}{r!}. \quad (34)$$

They are particularly natural for a Poisson limit: a Poisson variable of mean λ has $\kappa_1^{(F)} = \lambda$ and all higher factorial cumulants zero.

Theorem 7.1 (First four factorial cumulants). *Let $N = kn$. Whenever the displayed denominators are nonzero,*

$$\kappa_1^{(F)} = k - 1, \quad (35)$$

$$\kappa_2^{(F)} = -\frac{(k-1)^2}{N-1}, \quad (36)$$

$$\kappa_3^{(F)} = \frac{2(k-2)(k-1)^2}{(N-2)(N-1)}, \quad (37)$$

$$\kappa_4^{(F)} = -\frac{6(k-1)^2\{N(k^2-6k+7)+4k-6\}}{(N-3)(N-2)(N-1)^2}. \quad (38)$$

In particular,

$$\boxed{\mathbb{E}X_{n,k} = k - 1, \quad \text{Var } X_{n,k} = (k-1) - \frac{(k-1)^2}{kn-1}.} \quad (39)$$

Proof. Keep terms through v^4 in (20). Since

$$R_k(z) = 1 + a_{k,1}z + a_{k,2}z^2 + a_{k,3}z^3 + a_{k,4}z^4 + O(z^5),$$

ordinary multinomial expansion gives the first four factorial moments as rational functions of n and N . Taking the formal logarithm and using (14) simplifies to (35)–(38). This is finite polynomial algebra; Appendix A records a reproducible symbolic check. For any integer-valued random variable, $\mathbb{E}X = \kappa_1^{(F)}$ and $\text{Var } X = \kappa_1^{(F)} + \kappa_2^{(F)}$, which gives (39). $\square$

All higher factorial cumulants tend to zero. The all-orders theorem gives the sharper scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ for fixed $r \geq 2$, a mod-Poisson pattern that controls local corrections.

8 Explicit expansion through third order

Put $\lambda = k - 1$. Expanding the exact cumulants and applying Theorem 6.2 gives the following compact logarithmic form.

Theorem 8.1 (Logarithmic factorial-PGF expansion). *For fixed $k \geq 2$, uniformly for v in compact subsets of $\mathbb{C}$,*

$$\begin{aligned} \log \Phi_{n,k}(v) = & \lambda v - \frac{\lambda^2 v^2}{2kn} \\ & + \frac{\lambda^2 v^2 \{2(k-2)v - 3\}}{6k^2 n^2} \\ & - \frac{\lambda^2 v^2 \{(k^2 - 6k + 7)v^2 - 4(k-2)v + 2\}}{4k^3 n^3} + O_{k,v}(n^{-4}). \end{aligned} \quad (40)$$

Proof. Insert (36)–(38) into (34) and expand their denominators in $1/n$. These terms supply every contribution through n^{-3} because the fifth and higher factorial cumulants begin at $O_k(n^{-4})$, as follows from Theorem 6.2. Collecting powers of v gives (40). $\square$

Setting $v = -1$ produces the no-adjacency probability.

Corollary 8.2 (No-adjacency transseries). *For fixed $k \geq 2$,*

$$\log p_{n,k} = -(k-1) - \frac{(k-1)^2}{2kn} - \frac{(k-1)^2(2k-1)}{6k^2n^2} - \frac{(k-1)^4}{4k^3n^3} + O_k(n^{-4}), \quad (41)$$

$$p_{n,k} = e^{-(k-1)} \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2n^2} - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3n^3} + O_k(n^{-4}) \right]. \quad (42)$$

The logarithmic form is often numerically superior because exponentiating a truncated log preserves positivity and partially resums products of lower orders.

8.1 A first local correction

The expansion describes more than the mass at zero.

Corollary 8.3 (Fixed local probabilities). *For each fixed $j \geq 0$,*

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right]. \quad (43)$$

Proof. Exponentiating the first correction in (40) gives

$$\Phi_{n,k}(v) = e^{\lambda v} \left(1 - \frac{\lambda^2 v^2}{2kn} + O(n^{-2}) \right).$$

Write $v = u - 1$ and extract the coefficient of u^j . Dividing the correction by the Poisson mass gives

$$-\frac{\lambda^2}{2kn} \left(1 - \frac{2j}{\lambda} + \frac{j(j-1)}{\lambda^2} \right),$$

which is (43). $\square$

9 Specializations to the OEIS rows

For $k = 2, 3, 4, 5$, (42) becomes

$$p_{n,2} = e^{-1} \left(1 - \frac{1}{4n} - \frac{3}{32n^2} - \frac{1}{384n^3} + O(n^{-4}) \right), \quad (44)$$

$$p_{n,3} = e^{-2} \left(1 - \frac{2}{3n} - \frac{4}{27n^2} + \frac{4}{81n^3} + O(n^{-4}) \right), \quad (45)$$

$$p_{n,4} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right), \quad (46)$$

$$p_{n,5} = e^{-4} \left(1 - \frac{8}{5n} + \frac{8}{25n^2} + \frac{128}{375n^3} + O(n^{-4}) \right). \quad (47)$$

Thus (44) sharpens the known e^{-1} law for A007060 [2], (45) sharpens the e^{-2} law printed on A193624 [3], and (46) supplies explicit corrections for A330266 [1].

9.1 The standard 52-card deck

For $n = 13$ and $k = 4$, exact enumeration gives

$$p_{13,4} = 0.04547628233109430448968591097164118590821040881636 \dots \quad (48)$$

The successive approximations are

approximation	numerical value
e^{-3}	0.04978706836786394298
through n^{-1} in (46)	0.04547857206679879407
through n^{-2} in (46)	0.04547166742529067684
through n^{-3} in (46)	0.04547585004466578632
exponential of (41) through n^{-3}	0.04547622509184257975
exact	0.04547628233109430449

The third-order logarithmic approximation has relative error about 1.26×10^{-6} even at the modest value $n = 13$.

The exact mean number of equal-rank adjacencies is 3, while

$$\text{Var } X_{13,4} = 3 - \frac{9}{51} = \frac{48}{17} = 2.8235294117 \dots$$

This quantifies the slight underdispersion relative to a Poisson(3) law.

9.2 Numerical convergence across several multiplicities

The following table lists $e^{k-1}p_{n,k}$, which tends to 1. The final column is the signed error after subtracting all three displayed correction terms in (42).

k	n	$e^{k-1}p_{n,k}$	third-order residual
2	10	0.97406694843814	7.05×10^{-6}
2	50	0.99496248890932	9.74×10^{-9}
3	10	0.93191894503056	1.77×10^{-5}
3	50	0.98660782736509	2.49×10^{-8}
4	10	0.88747575415541	2.56×10^{-5}
4	50	0.97749213853337	3.70×10^{-8}
5	10	0.84356657118181	2.52×10^{-5}
5	50	0.96813076805226	3.74×10^{-8}
6	10	0.80107140086354	1.29×10^{-5}
6	50	0.95869552482316	1.94×10^{-8}

The small residuals are not used as proof; they are independent diagnostics of the symbolic expansion.

10 Asymptotics of the counts, ratios, and inverse functions

10.1 Logarithmic count transseries

Let $N = kn$ and $\lambda = k - 1$. Combining Stirling's expansion with (41) yields

$$\log B_{n,k} = N(\log N - 1) + \frac{1}{2} \log(2\pi N) - \lambda + \frac{A_1}{N} + \frac{A_2}{N^2} + \frac{A_3}{N^3} + O_k(N^{-4}), \quad (49)$$

where

$$A_1 = \frac{1}{12} - \frac{\lambda^2}{2}, \quad A_2 = -\frac{\lambda^2(2k-1)}{6}, \quad A_3 = -\frac{1}{360} - \frac{\lambda^4}{4}. \quad (50)$$

For multiset words,

$$\log S_{n,k} = \log B_{n,k} - n \log(k!). \quad (51)$$

These formulas separate the universal factorial growth from the finite adjacency penalty.

10.2 Successive-term ratios

For fixed k ,

$$\frac{B_{n+1,k}}{B_{n,k}} = \prod_{j=1}^k (kn + j) \left[1 + \frac{(k-1)^2}{2kn^2} + \frac{(k-1)^2(k-2)}{6k^2n^3} + O_k(n^{-4}) \right]. \quad (52)$$

Indeed, the factorial ratio is exact and subtraction of $\log p_{n,k}$ from $\log p_{n+1,k}$ gives the bracket. Such a ratio is useful for checking recurrences attached to individual OEIS rows.

10.3 A Lambert- W inverse transseries

Suppose a large value Y is known and one wants to estimate the index n for which $B_{n,k} \approx Y$. Set

$$L = \log Y + \lambda, \quad N_0 = \frac{L}{W_0(L/e)}. \quad (53)$$

Then $N_0(\log N_0 - 1) = L$. One Newton correction from the remaining terms in (49) gives

$$N = N_0 - \frac{\frac{1}{2} \log(2\pi N_0) + A_1/N_0 + A_2/N_0^2 + A_3/N_0^3}{\log N_0} + O_k(N_0^{-1}), \quad n = \frac{N}{k}. \quad (54)$$

This is an inverse transseries in the same spirit as the inversion examples in the ProveIt repository. Further Newton steps and further terms of Theorem 6.2 extend it mechanically. [Added 1 October 2026, batch 7302: the inversion examples meant are those of the canonical volume *Transseries and Inversion* [13]. The forward map $\log B_{n,k}$ of (49) is the volume's gamma carrier (theorem "All-orders expansion of Γ_+^{-1} ", label `p6:thm:gamma`) perturbed by the power series in $1/N$ of (41), the situation of its theorem "Perturbed inversion" (`p0:thm:perturbed-inversion`); the volume's chapter "The subfactorial" (`p8:sec:top`) treats the derangement numbers the same way. Normalizations differ: here $L = \log Y + \lambda$ and $N_0 = L/W_0(L/e)$, with $\frac{1}{2} \log(2\pi N_0)$ left in the correction; the volume uses $R = \log y - \frac{1}{2} \log(2\pi)$. Formula (54) is a first-order instance of that apparatus, and no novelty is claimed for it. Deciding the integer index from an approximate inverse, as in the deck example

below and in research question 11 of Section 14, is the subject of the volume’s staircase theorem (`p0:thm:staircase`).]

For the exact standard-deck count in (48), with $k = 4$, the leading Lambert estimate gives $n \approx 13.17720934$. The correction (54) gives

$$n \approx 12.99979618,$$

which correctly identifies the integer index 13.

11 Relation to rook polynomials and recent multipartite work

The coefficients $a_{k,m} = m! \binom{k}{m} \binom{k-1}{m}$ are the familiar rook numbers of a rectangular board of dimensions k by $k-1$. This explains why rook-polynomial methods naturally occur in exact deck-shuffle enumeration. Enouen’s report on the standard deck [8] develops such a direct rook-polynomial approach. The reciprocal polynomial $C_k(z) = R_k(-z)$ packages all ranks into the compact power $C_k(z)^n$.

Balanced Smirnov words have also been connected explicitly with Hamiltonian paths and cycles in complete multipartite graphs. Recent work by Mehiri [9] gives bijections and inclusion–exclusion formulas, including nonuniform part sizes. The present asymptotic regime keeps the part size k fixed and lets the number of parts n grow. In this regime, Theorem 5.1 says:

Among all ordered vertex lists of $K_{k,k,\dots,k}$, the fraction that form Hamiltonian paths tends to $e^{-(k-1)}$, and the number of forbidden within-part consecutive steps tends to Poisson($k-1$).

This probabilistic statement complements exact Hamiltonian enumerations and suggests analogous cyclic and nonuniform limit laws.

12 Reproducibility and a formalization route

The accompanying program `code/verify.py` performs the following tasks using exact Python integers for enumeration and high-precision decimal arithmetic for probabilities:

1. constructs $C_k(z) = \sum_m (-1)^m a_{k,m} z^m$;
2. raises it to the n th power by exact polynomial convolution;
3. evaluates (18);
4. checks the initial terms of A007060, A193624, and A330266;
5. verifies the exact 52-card value;
6. writes the numerical convergence table used above.

The optional SymPy script `code/derive_symbolic.py` reconstructs the first four factorial cumulants and checks the displayed n^{-3} expansion. The article does not rely on a fitted numerical constant.

A Lean formalization can be separated into finite and analytic layers. The finite core is especially approachable:

1. define rank words, runs, and the statistic X ;
2. prove the run-compression bijection and finite coefficient identity (8);
3. prove the polynomial identity (11);
4. reduce the integral notation to the factorial moment identity $\int_0^\infty t^m e^{-t} dt = m!$;
5. prove the exact finite sum (18);
6. establish the leading factorial-moment estimate using polynomial degree bounds.

The final method-of-moments step can either use an existing probability convergence library or be replaced by direct convergence of each fixed mass through the probability generating function. The exact count identities can be formalized independently of the asymptotic layer.

13 Proposed OEIS updates

Subject to independent review and attribution, the following concise comments would strengthen the relevant entries.

General fixed- k family. If $B_{n,k}$ counts permutations of kn labeled cards, k in each of n ranks, with no equal adjacent ranks, then for each fixed $k \geq 2$,

$$\frac{B_{n,k}}{(kn)!} = e^{-(k-1)} \left(1 - \frac{(k-1)^2}{2kn} + O_k(n^{-2}) \right).$$

More precisely, the number of equal-rank adjacencies converges to $\text{Poisson}(k-1)$, and (42) gives three correction terms.

A330266. Add (46), together with the exact variance $3 - 9/(4n-1)$ for the adjacency statistic. The integral on the entry is the $k=4$ instance of

$$B_{n,k} = \int_0^\infty \{(-1)^k k! L_k^{(-1)}(t)\}^n e^{-t} dt.$$

A007060 and A193624. Add (44) and (45), respectively. Their multiset partners A114938 and A193638 have the same normalized probabilities because of (3).

A plain-text version is included as `oeis_proposed_updates.txt`; it is explicitly marked as unsubmitted. [Added 1 October 2026, batch 73O2: shipped as `data/oeis_proposed_updates.txt`, unchanged. Nothing has been submitted to the OEIS. Its correction terms rest on the sketched step of Remark 6.3.]

14 Further research questions and topics

1. **Uniformly growing multiplicity.** Determine the largest range $k = k(n)$ in which a Poisson approximation remains valid, and locate transitions to compound-Poisson or normal behavior. The fixed- k proof is not uniform when the degree of R_k grows.

2. **Explicit total-variation bounds.** Use Stein–Chen coupling or the gamma representation to obtain a sharp bound of the form $d_{\text{TV}}(X_{n,k}, \text{Poisson}(k-1)) \leq C_k/n$, preferably with the optimal constant and a computable second-order signed correction.
3. **Cyclic words and Hamiltonian cycles.** Also forbid equality between the last and first ranks. Find the exact analogue of (40), taking proper account of oriented versus unoriented cycles and rotational symmetries.
4. **Unequal multiplicities.** For ranks of sizes $k_1, \dots, k_n$ and total N , identify natural sparse conditions under which the adjacency count is asymptotically Poisson with mean $\sum_i k_i(k_i-1)/N$. Determine the correction terms in symmetric sums of the k_i .
5. **Larger exclusion distance.** Forbid two equal ranks within distance d rather than only distance one. Cluster expansions should produce a Poisson or compound-Poisson law when k, d are fixed, but overlapping short patterns require new local cumulants.
6. **Moderate and large deviations.** Analyze $\mathbb{P}(X_{n,k} \geq x)$ outside the fixed- x regime. The exact gamma expectation may lead to a saddle-point approximation uniformly for x growing with n .
7. **All-order closed forms.** The operator rule (33) is constructive. Find a canonical Bell-polynomial expression for the coefficient of n^{-r} in $\log \Phi_{n,k}(v)$, and study sign or factorization patterns in k and v .
8. **Laguerre-zero geometry.** Clarify how the real zeros of $L_k^{(-1)}$ govern recurrences, cancellations, and possible saddle points of the exact integral. The integral involves a sign-changing polynomial, so a naive positive Laplace principle is insufficient.
9. **Differential equations and recurrences.** For each fixed k , derive minimal-order polynomial recurrences for $B_{n,k}$ and compare their asymptotic solutions with (42). Seek bounds on recurrence order as a function of k .
10. **New OEIS rows and arrays.** Create a two-dimensional array for $S_{n,k}$ or $B_{n,k}$, with rows indexed by k , including $k = 5, 6$ and exact kernels P_k . Cross-reference the labeled, multiset, rook-polynomial, and multipartite-graph interpretations.
11. **Rigorous inverse error bounds.** Turn (54) into an explicit bracketing theorem for the least n with $B_{n,k} \geq Y$, including monotonicity thresholds and certified rounding.
12. **Machine-checked asymptotics.** Formalize the finite identities in Lean, then isolate a reusable theorem that turns fixed factorial-moment convergence plus tightness into Poisson convergence. This would support many related OEIS limit laws.

15 Conclusion

The constant $e^{-(k-1)}$ is not an isolated feature of three OEIS rows. It is the zero mass of a limiting Poisson law for the complete adjacency statistic. The run transform converts the combinatorics into a single Laguerre polynomial; reciprocal coefficients give exact inclusion–exclusion; and a gamma negative-moment representation produces systematic correction terms. The resulting framework simultaneously proves the fixed- k conjecture recorded on A330266, sharpens the known $k = 2, 3, 4$ asymptotics, and opens a tractable family of cyclic, nonuniform, and growing-multiplicity problems.

A Cumulant algebra

For completeness, write

$$R_k(z) = 1 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + O(z^5),$$

where

$$\begin{aligned} a_1 &= k(k-1), \\ a_2 &= \frac{k(k-2)(k-1)^2}{2}, \\ a_3 &= \frac{k(k-3)(k-2)^2(k-1)^2}{6}, \\ a_4 &= \frac{k(k-4)(k-3)^2(k-2)^2(k-1)^2}{24}. \end{aligned}$$

The coefficient of v^r in (20) is $[z^r]R_k(z)^n/(N)_{\underline{r}}$. Thus, for example,

$$\begin{aligned} \mathbb{E}(X)_{\underline{2}} &= \frac{2na_2 + n(n-1)a_1^2}{(N)_{\underline{2}}}, \\ \mathbb{E}(X)_{\underline{3}} &= \frac{6na_3 + 6n(n-1)a_2a_1 + n(n-1)(n-2)a_1^3}{(N)_{\underline{3}}}. \end{aligned}$$

The fourth factorial moment is obtained from the five partitions 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1. Applying the standard moment-to-cumulant relations and simplifying gives (35)–(38). The accompanying SymPy script performs exactly this finite calculation and verifies that the difference from (40) is $O(n^{-4})$.

B Exact initial terms for $k = 5$

The same formula supplies a useful prospective OEIS row. For $n = 0, 1, \dots, 7$,

$$\begin{aligned} B_{n,5} &= 1, \ 0, \ 28800, \ 12420864000, \ 27917203599360000, \\ &\quad 1977269653324800000000000, \\ &\quad 3614614673967242167910400000000, \\ &\quad 147444323126376251733903684403200000000. \end{aligned}$$

The kernel is $P_5(t) = t^5 - 20t^4 + 120t^3 - 240t^2 + 120t$, and $B_{n,5} = \int_0^\infty P_5(t)^n e^{-t} dt$. These terms should be independently cross-checked before creating or editing an OEIS entry.

C Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 57 of batch 73 of ProveIt’s incoming-reports intake: the archive `ProveIt_Balanced_Smirnov_OEIS.zip` (main file `article.tex`, a 17-page PDF). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted

the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision and cites no repository file. It has no author or preparer line: its author field is the subtitle “A research note on OEIS A007060, A193624, A330266, and related sequences”, and the delivered PDF’s author metadata is empty.

What the write changed.

- Every label gained the prefix `bsw::`; none was otherwise renamed or removed. The manuscript had 66 labels.
- Citation commands were added: the manuscript listed twelve bibliography entries but cited none of them. They now stand at the first mentions of the OEIS entries, the Smirnov-word generating function, the Laguerre expansion, and the works of Eriksson–Martin, Enouen and Mehiri.
- The authors and dates of the two OEIS contributions are named: S. Kirgizov’s fixed- k conjecture (29 September 2023) and D. Radcliffe’s deduction of the case $k = 4$ (9 September 2025).
- Added: the editorial note after the scope paragraph, Remark 6.3 on the sketched analytic step, a note in the inverse section citing the transseries volume [13], a note on the shipped file of proposed OEIS comments, one bibliography entry, the label `bsw:sec:questions`, and this section.
- Nothing was removed. Every non-claim of the manuscript is kept: no complete literature search or priority claim, the numerical residuals “not used as proof”, the cross-check required for the $k = 5$ terms, and the unsubmitted status of the proposed OEIS comments.

Relation to the companion report. The path-forest report `a189281-path-forest-expansions` of this collection (batch 73, manuscript 60) follows the same chain of method (exact marked-subset generating function, Poisson limit, all-orders $1/n$ expansion of a factorial probability generating function, Lambert- W inversion) for permutations avoiding a fixed positional and value offset, with limit Poisson(θ), $\theta \in \{1, 2\}$. Neither theorem specializes to the other. In the present model the n ranks give n cliques K_k as the “value” relation, whose number grows with n , so the hypotheses of the path-forest report (a fixed number of long paths) do not hold. Its uniform factorial-moment bound is the device named in Remark 6.3.

Checks by the intake. The intake reran both shipped programs on a copy (the CSV is byte-identical; the deck output equal apart from line endings; the symbolic script confirms $\kappa_1^{(F)}, \dots, \kappa_4^{(F)}$ and the n^{-3} expansion), compared the terms of A007060, A193624 and A330266 with an independent dynamic program over rank words, confirmed the $k = 5$ terms (for $n \leq 4$ by that program, for $n \leq 7$ by the formula) and the 52-card value, checked the variances $4/5, 6/5, 6/7$ at $(n, k) = (3, 2), (2, 3), (4, 2)$ by full enumeration, and rederived by hand the local correction, the ratio bracket and A_1, A_2, A_3 . These checks do not replace an independent review of the proofs.

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- [13] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, canonical volume of the ProveIt series-and-transseries collection. Its source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (labels `p0:thm:perturbed-inversion`, `p0:thm:staircase`), chapter on the increasing inverse of Gamma (`p6:thm:gamma`) and chapter “The subfactorial” (`p8:sec:top`). Repository state of placement commit `6e193dd4f`.